



REPUBLIC OF KENYA

COMPETENCY BASED CURRICULUM

FOR

BUILDING TECHNOLOGY

KNQF LEVEL 5

(CYCLE 3)

ISCED PROGRAMME CODE: 0732 454 A



TVET CDACC
P.O. BOX 15745-00100
NAIROBI

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FOREWORD

The provision of quality education and training is fundamental to the Government's overall strategy for social and economic development. Quality education and training contribute to the achievement of Kenya's development blueprint and sustainable development goals.

Reforms in the education sector are necessary to achieve Kenya Vision 2030 and meet the provisions of the Constitution of Kenya 2010. The education sector had to be aligned to the Constitution, and this resulted in the formulation of the Policy Framework for Reforming Education and Training in Kenya (Sessional Paper No. 14 of 2012). A key feature of this policy is the radical change in the design and delivery of TVET training. This policy document requires that training in TVET be competency-based, curriculum development be industry-led, certification be based on demonstration of competence, and the mode of delivery allow for multiple entry and exit in TVET programmes.

These reforms demand that Industry takes a leading role in curriculum development to ensure the curriculum addresses its competence needs. It is against this background that this curriculum has been developed. For trainees to build their skills on foundational hands-on activities of the occupation, units of learning are grouped in modules. This has eliminated duplication of content and streamlined exemptions based on skills acquired as a trainee progresses in the up-skilling process, while at the same time allowing trainees to be employable in the shortest time possible through the acquisition of part qualifications.

It is my conviction that this curriculum will play a great role in developing competent human resources for the Construction Sector's growth and development.

PRINCIPAL SECRETARY

STATE DEPARTMENT FOR TVET

MINISTRY OF EDUCATION

PREFACE

Kenya Vision 2030 aims to transform Kenya into a newly industrializing middle-income country, providing high-quality life to all its citizens by the year 2030. Kenya intends to create globally competitive and adaptive human resource base to meet the requirements of a rapidly industrializing economy through lifelong education and training. TVET has a responsibility to facilitate the process of inculcating knowledge, skills, and worker behaviour necessary for catapulting the nation to a globally competitive country, hence the paradigm shift to embrace Competency-Based Education and Training (CBET).

TVET ACT, CAP 210A and Sessional Paper No. 1 of 2019 on Reforming Education and Training in Kenya for Sustainable Development emphasized the need to reform curriculum development, assessment, and certification. This called for a shift to CBET to address the mismatch between skills acquired through training and skills needed by industry, as well as increase the global competitiveness of the Kenyan labour force.

This curriculum has been developed in adherence to the Kenya National Qualifications Framework and CBETA standards and guidelines. The curriculum is designed and organized into Units of Learning with Learning Outcomes, suggested delivery methods, learning resources, and methods of assessing the trainee's achievement. In addition, the units of learning have been grouped in modules to concretize the skills acquisition process and streamline upskilling.

I am grateful to all expert trainers and everyone who played a role in translating the Occupational Standards into this competency-based modular curriculum.

**CHAIRMAN
TVET CDACC**

ACKNOWLEDGEMENTS

This curriculum has been designed for competency-based training and has independent units of learning that allow the trainee flexibility in entry and exit. In developing the curriculum, significant involvement and support were received from expert trainers, institutions and organizations.

I recognize with appreciation the role of the National Sector Skills Committee (NSSC) in ensuring that competencies required by the industry are addressed in the curriculum. I also thank all stakeholders in the construction sector for their valuable input and everyone who participated in developing this curriculum.

I am convinced that this curriculum will go a long way in ensuring that individuals aspiring to work in the construction Sector acquire competencies to perform their work more efficiently and effectively.

COUNCIL SECRETARY/CEO

TVET CDACC

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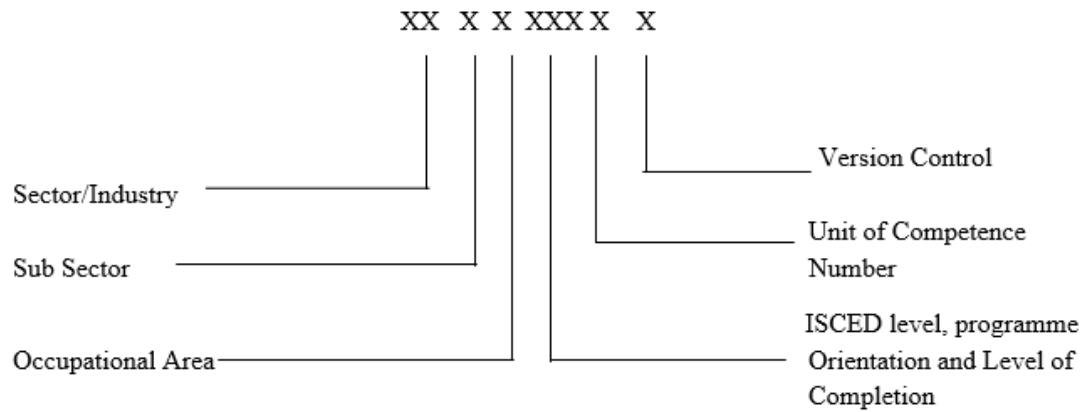
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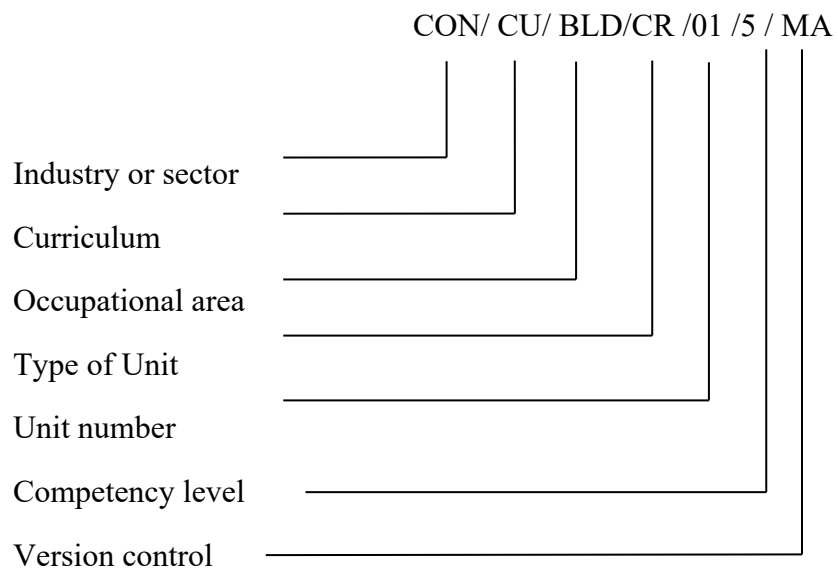
ABBREVIATIONS AND ACRONYMS

BRC	- BRITISH REINFORCEMENT CONCRETE
CBET	- COMPETENCY BASED EDUCATION AND TRAINING
CD	-COMPACT DISK
CPU	- CENTRAL PROCESSING UNIT
CV	- CURRICULUM VITAE
DPM	- DAMP PROOF MEMBRANE
DVD	-DIGITAL VERSATILE DISK
DVI	-DIGITAL VISUAL INTERFACE
EMCA	-ENVIRONMENTAL MANAGEMENT AND COORDINATION ACT
HDMI	-HIGH DEFINATION MULTIMEDIA INTERFACE
ICT	-INFORMATION COMMUNATION TECHNOLOGY
IOT	-INTERNET OF THINGS
MITM	-MAN IN THE MIDDLE
PHD	-DOCTOR OF PHILOSOPHY
PPEs	-PERSONAL PROTECTIVE EQUIPMENTS
TVET	-TECHNICAL VOCATIONAL EDUCATION AND TRAINING
URI	-UNIFORM RESOURCE IDENTIFIER
USB	-UNIVERSAL SERIAL BUS
VGA	-VIDEO GRAPHICS ARRAY
Dr.	-DOCTOR

KEY TO ISCED UNIT CODE



KEY TO TVET CDACC UNIT CODE



COURSE OVERVIEW

Building Technology Level 5 qualification consist of competencies that a person must acquire to enable him/her to be certified as a Building Technology Craftsperson. The competencies include performing masonry structures construction, performing masonry work finishes, producing of simple masonry products, construction site preliminary works, executing building substructure, superstructure, building finishes and external works. The units of Learning comprising Building Technician Level 5 include the following basic, common and core units:

SUMMARY OF UNITS OF LEARNING

MODULE I

ISCED Unit Code	TVET CDACC UNIT CODE	Units Title	Unit Duration (Hours)	Credit Factor
0732 251 01A	CON/CU/BLD/CR/01/3/MA	Masonry works	130	13.0
0732 251 02A	CON/CU/BLD/CR/02/3/MA	Masonry Works Finishes I	90	9.0
0732 251 03A	CON/CU/BLD/CR/03/3/MA	Masonry Unit Production	100	10.0
	SUB TOTAL		320	32.0

MODULE II

ISCED Unit Code	TVET CDACC UNIT	Units Title	Unit Duration (Hours)	Credit Factor
0732 351 04A	CON/CU/BLD/CR/01/4/MA	Building Site Preliminary Works I	120	12.0
0732 351 05A	CON/CU/BLD/CR/02/4/MA	Building Temporary Works	120	12.0
		SUB TOTAL	240	24.0

MODULE III

0611 451 06 A	CON/CU/BLD/BC/01/5/MA	Digital Literacy	40	4.0
0031 451 07A	CON/CU/BLD/BC/02/5/MA	Communication skills	40	4.0
541 451 08A	CON/CU/BLD/CC/01/5/MA	Basic mathematics I	60	6.0
0732 451 09A	CON/CU/BLD/CC/02/5/MA	Technical drawing	80	8.0
0732 451 10A	CON/CU/BLD/CR/01/5/MA	Building Substructure works	120	12.0
0732 451 11A	CON/CU/BLD/CR/02/5/MA	Doors and windows	50	5.0
0732 451 12A	CON/CU/BLD/CR/03/5/MA	External works	60	6.0
		SUBTOTAL	450	45.0

MODULE IV

0417 451 13A	CON/CU/BLD/BC/03/5/MA	Work Ethics and Practices	40	4.0
0413 451 14A	CON/CU/BLD/BC/04/5/MA	Entrepreneurial skills	40	4.0
0541 451 15A	CON/CU/BLD/CC/03/5/MA	Basic mathematics II	80	8.0
0732 451 16A	CON/CU/BLD/CC/04/5/MA	Construction material science	60	6.0
0732 451 17A	CON/CU/BLD/CR/04/5/MA	Building superstructure work	80	8.0
0732 451 18A	CON/CU/BLD/CR/05/5/MA	Building Works Finishes II	80	8.0
		SUBTOTAL	380	38.0
		Industry Training	480	
		GRAND TOTAL	1870	187.0

Entry Requirements

An individual entering this course should have any of the following minimum requirements:

- a) Kenya Certificate of Secondary Education (KCSE) grade D

Or

- b) Masonry Level 4 Certificate

Or

- c) Equivalent KNQF level 4 qualification as determined by relevant regulatory body

Trainer qualifications

Qualifications of a trainer for this course include:

- Possession of a higher qualification than building technology level 5 or in related trade area; and
- Be registered by TVETA

iii) Be Licensed by Engineering Board of Kenya

Industry Placement/Training

An individual enrolled in this course will be required to undergo Industrial attachment for a minimum period of 480 hours in a construction sector. The industrial training may be taken after completion of all units for those pursuing the full qualification or be distributed equally in each unit for those pursuing part qualification. In the case of dual training model, industrial training shall be as guided by the dual training policy.

Assessment

The course shall be assessed formatively and summatively:

- a) During formative assessment all performance criteria shall be assessed based on performance criteria weighting.
- b) Summative assessment shall focus more on critical aspects of the Unit of competency.
- c) During summative assessment basic and common units shall be integrated or assessed concurrently with the core units.
- d) Formative and summative assessment weights shall constitute 60% and 40% of the overall score respectively.
- e) Theoretical and practical weight for each unit of learning shall be as follows;
 - i) 10:90 for units in module I and module II
 - ii) 30:70 for units in module III and module IV
- f) For a candidate to be declared competent in a unit of competency, the candidate must meet the following conditions:
 - i) Obtained at least 40% in theory assessment in formative and summative assessments.
 - ii) Obtained at least 50% in practical assessment in formative and summative assessment where applicable.
 - iii) Obtained at least 50% in the weighted results between formative assessment and summative assessment where the former constitutes 60% and the latter 40% of the overall score.
- g) Assessment performance rating for each unit of competency shall be as follows:

MARKS	COMPETENCE RATING
80 -100	Mastery
65 – 79	Proficiency
50 – 64	Competent
49 and below	Not Yet Competent
Y	Assessment Malpractice/irregularities

- h) Assessment for Recognition of Prior Learning (RPL) may lead to award of part and/or full qualification

Certification

A candidate will be issued with a Certificate of Competency upon demonstration of competence in a core Unit of Competency. To be issued with Kenya National TVET Certificate in Building Technology Level 6, the candidate must demonstrate competence in all the Units of Competency as given in the qualification pack. A Statement of Attainment certificate may be issued upon demonstration of competence in a certifiable element within a unit.

The certificates will be issued by the TVET CDACC

MODULE I

MASONRY WORKS

UNIT CODE: 0732 251 01A

TVET CDACCUNIT CODE: CON/CU/BLD/CR /01/3/MA

UNIT DURATION: 130 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform Masonry Works

Unit Description

This unit specifies the competencies required to perform masonry works. It entails Interpret building drawings, setting out the building, laying of foundation concrete, construction of foundation walls, Construct ground floor slab, Construct reinforced concrete elements and Construct superstructure walls. It applies in the construction industry.

Summary of Learning Outcomes:

Learning Outcomes	Duration (Hours)
1. Interpret building drawings	10
2. Set-out building	20
3. Lay building foundation concrete	20
4. Construct foundation walls	20
5. Construct ground floor slab	30
6. Construct reinforced concrete elements	10
7. Construct superstructure walls	20
TOTAL	130

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods

1. Interpret Building drawings	1.1 Symbols and abbreviation 1.2 Measurements 1.3 Building elevations 1.4 Building sections	• Practical • Projects • Third Party Reports • Written tests
2. Set-out building	2.1 PPEs 2.1.1 Types 2.1.2 uses 2.2 Setting out tools and equipment's 2.2.1 Types 2.2.2 uses 2.3 Setting out methods 2.3.1 3-4-5 Method 2.3.2 Builder's square 2.4 Profiling 2.4.1 Profile boards 2.5 Levelling 2.5.1 Surface levelling 2.6 Transferring measurements to the ground	• Observation • Written tests • Oral questioning • Third party report • Portfolio
3. Lay building foundation concrete	3.1 PPEs 3.1.1 Types 3.1.2 uses 3.2 Materials and tools 3.2.1 Types 3.2.2 uses 3.3 Excavation of trenches 3.3.1 Methods 3.3.1.1 Mechanical	• Practical • Projects • Third Party Reports • Written tests

	3.3.1.2 Manual 3.3.1.3 Procedures 3.4 Foundation trenches 3.4.1 Types 3.4.2 Strip 3.4.3 Pad foundations 3.5 Foundation treatment 3.5.1 procedures 3.6 Foundation blinding 3.6.1 procedures	
4. Construct foundation walls	4.1 Measurement transfer 4.1.1 Procedures 4.2 Foundation walling units 4.2.1 Types 4.2.1.1 Masonry blocks 4.2.1.2 Bricks 4.2.1.3 laying procedures 4.3 Backfilling 4.3.1 procedures	• Practical • Projects • Third Party Reports • Written tests
5. Construct ground floor slab	5.1 PPEs 5.1.1 Types 5.1.2 Uses 5.2 Hardcore 5.2.1 laying procedures 5.3 Murram blinding 5.3.1 Laying procedures 5.4 Formwork 5.4.1 erection procedures 5.5 Damp proofing 5.5.1 Types 5.5.1.1 Damp proofing membrane 5.5.1.2 Damp proof spray 5.5.1.3 Waterproofing additives 5.5.2 procedures	• Practical • Projects • Third Party Reports • Written tests

	5.6 Concrete casting 5.6.1 procedures 5.7 Curing 5.7.1 methods 5.7.1.1 Water 5.7.1.2 Jute Sacks 5.7.1.3 Blankets 5.7.1.4 Sand 5.7.1.5 Curing agents 5.7.2 procedures	
6. Construct reinforced concrete elements	6.1 PPEs 6.1.1 Types 6.1.2 Uses 6.2 Column concrete kicker 6.2.1 Casting 6.2.2 Curing 6.3 Concrete elements 6.3.1 Casting 6.3.1.1 Columns 6.3.1.2 Beams 6.4 Curing	• Practical • Projects • Third Party Reports • Written tests
7. Construct superstructure walls	7.1 PPEs 7.1.1 Types 7.1.2 Uses 7.2 Damp Proof Course 7.2.1 Laying procedure 7.3 Masonry wall units 7.3.1 Mortar preparation 7.3.2 Laying procedure 7.4 Bridging of walls 7.4.1 Arches 7.4.2 Lintels	• Practical • Projects • Third Party Reports • Written tests

Suggested Methods of Instruction

- Practical
- Project
- Demonstrations
- Group Discussions
- Direct instructions

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Reference materials i.e. textbooks		14 pcs	1:2
2.	Rolls Flip Charts		5 pcs	1:5
B	Learning Facilities & infrastructure			
3.	Whiteboards		1 pc	1:25
4.	Lecture room	9m by 8m	1	
5.	Workshop	10m by 10m	1	
6.	Desktop computers/laptops		14 pcs	1:2
C	Consumable materials			
6.	Assorted colour of whiteboard markers		10 pcs	10:1
7.	Assorted timber		Sufficient	
8.	Assorted nails		Sufficient	
9.	Pegs		Sufficient	
10.	Masonry walling units		Sufficient	
11.	Damp proof course		Sufficient	
12.	Hoop iron		Sufficient	
13.	Mortar		Sufficient	
D	Tools and Equipment			
14.	Technical Drawing Instruments (Rulers,		25 Sets	1:1

	Protractors, Compasses, Set-squares)			
15.	Builders Line		5 pcs	1:5
16.	Mason Square		5 pcs	1:5
17.	Claw Hammer		5 pcs	1:5
18.	Claw bar		5 pcs	1:5
19.	Tape measure		5 pcs	1:5
20.	Spirit Level		5 pcs	1:5
21.	Mason Hammer		5 pcs	1:5
22.	Shovel		5 pcs	1:5
23.	Bucket		5 pcs	1:5
24.	Wheelbarrow		5 pcs	1:5
25.	Trowel		5 pcs	1:5
26.	Plumb Bob		5 pcs	1:5
27.	Wood Float		5 pcs	1:5

MASONRY WORKS FINISHES I

UNIT CODE: 0732 251 02A

TVET CDACC UNIT CODE: CON/CU/BLD/CR/02/3/MA

UNIT DURATION: 90 Hours

Relationship to Occupational Standards

This unit addresses the Unit of competency: Perform Masonry Works Finishes I

Unit Description

This unit specifies the competencies required to perform masonry works finishes. It entails Carrying out wall plastering, laying floor screed, carrying out Rendering and Carrying out keying and pointing. It applies in the construction industry.

Summary of Learning Outcomes

Learning Outcomes	Duration (Hours)
1. Carry out wall plastering	30
2. Lay floor screed	30
3. Carry out wall Rendering	15
4. Carry out Keying and Pointing	15
TOTAL	90

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Carry out wall plastering	1.1 Personal protective equipment 1.1.1 Types 1.1.2 Uses 1.2 Wall Plumbness 1.2.1 Checking procedure	• Practical • Projects • Third Party Reports • Written tests

	1.2.2 Dot guides 1.2.3 Plastering 1.2.4 Application procedure 1.3 Plaster curing	
2. Lay floor screed	2.1 Personal protective equipment 2.1.1 Types 2.1.2 Uses 2.2 Floor preparation procedure 2.3 Screeding 2.3.1 Types 2.3.1.1 Bonded Screed 2.3.1.2 Unbonded Screed 2.3.1.3 Floating Screed 2.3.2 Laying procedure 2.4 Screed curing	• Practical • Projects • Third Party Reports • Written tests
3. Carry out wall Rendering	3.1 Personal protective equipment 3.1.1 Types 3.1.2 Uses 3.2 Wall preparation Procedure 3.3 Rendering 3.3.1 Materials 3.3.1.1 Cement 3.3.1.2 Aggregates 3.3.2 Application procedure 3.4 Curing of render	• Practical • Projects • Third Party Reports • Written tests
4. Carry out keying and pointing.	4.1 Wall surface preparation 4.2 Mortar mix 4.3 Keying 4.4 Pointing 4.4.1 Tuck	• Practical • Projects • Third Party Reports • Written tests

	4.4.2 Flush	
	4.4.3 Recessed	
	4.4.4 Weathered	

Suggested Methods of Instruction

- Practical
- Project
- Demonstrations
- Group Discussions
- Direct instructions

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Reference materials i.e. textbooks		14 pcs	1:2
2.	Rolls Flip Charts		5 pcs	1:5
B	Learning Facilities & infrastructure			
3.	Whiteboards	For trainer's use	1 pc	1:25
4.	Lecture room	9m by 8m	1	
5.	Workshop	10m by 10m	1	
6.	Desktop computers/laptops	For both trainer's and trainee' use	25 pcs	1:1
C	Consumable materials			
6.	Assorted colour of whiteboard		10 pcs	10:1

	markers			
7.	Assorted timber		Sufficient	
8.	Assorted nails		Sufficient	
9.	Pegs		Sufficient	
10.	Mortar		Sufficient	
D	Tools and Equipment			
11.	Builders Line		5 pcs	1:5
12.	Mason Square		5 pcs	1:5
13.	Claw Hammer		5 pcs	1:5
14.	Claw bar		5 pcs	1:5
15.	Tape measure		5 pcs	1:5
16.	Spirit Level		5 pcs	1:5
17.	Mason Hammer		5 pcs	1:5
18.	Shovel		5 pcs	1:5
19.	Bucket		5 pcs	1:5
20.	Wheelbarrow		5 pcs	1:5
21.	Trowel		5 pcs	1:5
22.	Plumb Bob		5 pcs	1:5
23.	Wood Float		5 pcs	1:5
24.	Steel float		5 pcs	1:5
25.	Straight edge		5 pcs	1:5

26.	Keying tool		5 pcs	1:5
27.	Mason chisel		5 pcs	1:5
28.	Spade		5 pcs	1:5

MASONRY UNITS PRODUCTION

UNIT CODE: 0732 251 03A

TVET CDACC UNIT CODE: CON/CU/BLD/CR/03/3/MA

UNIT DURATION: 100 Hours

Relationship to Occupational Standards

This unit addresses the Unit of competency: Produce Masonry Products

Unit Description

This unit specifies the competencies required to produce simple masonry products. It entails Interpret Masonry units' drawings, production of masonry clay units, concrete masonry products and stabilized soil masonry products and performing stone dressing.

Summary of Learning Outcomes

Learning Outcomes	Duration (Hours)
1. Interpret masonry units' drawings	10
2. Produce clay masonry units	20
3. Produce concrete masonry products	30
4. Produce stabilized soil masonry units	20
5. Perform stone dressing	20
TOTAL	100

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Interpret Masonry units' drawings	1.1 Dimensions 1.2 Elevations 1.3 Measuring angles	• Practical • Projects • Third Party Reports • Written tests

2. Produce clay masonry units	2.1 Personal protective equipment 2.1.1 Types 2.1.2 Uses 2.2 Clay preparation 2.2.1 Procedure 2.3 Clay bricks 2.3.1 Moulding procedure. 2.3.2 Sun drying 2.4 Kilning 2.4.1 Procedure	• Practical • Projects • Third Party Reports • Written tests
3. Produce concrete masonry products	3.1 Personal protective equipment 3.1.1 Types 3.1.2 Uses 3.2 Concrete 3.2.1 Block specification 3.2.2 Preparation procedure 3.3 Concrete blocks curing	• Practical • Projects • Third Party Reports • Written tests
4. Produce stabilized soil masonry units	4.1 Personal protective equipment 4.1.1 Types 4.1.2 Uses 4.2 Materials 4.2.1 Job specifications 4.2.2 Preparation procedure 4.3 Soil blocks 4.3.1 Preparation procedure 4.4 Soil blocks curing	• Practical • Projects • Third Party Reports • Written tests
5. Perform stone dressing	5.1 PPEs 5.1.1 Types 5.1.2 Uses 5.2 Stone dressing tools and equipment 5.2.1 Types 5.2.2 Uses	• Practical • Projects • Third Party Reports • Written tests

	5.2.3 Stone dressing	
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Suggested Methods of Instruction

- Practical
- Project
- Demonstrations
- Group Discussions
- Direct instructions

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Reference materials i.e. textbooks	For trainee's use	14 pcs	1:2
2.	Rolls Flip Charts	For both trainer's and trainee' use	5 pcs	1:5
B	Learning Facilities & infrastructure			
3.	Whiteboards	For trainer's use	1 pc	1:25
4.	Lecture room	9m by 8m	1	
5.	Workshop/shade	10m by 10m	1	
6.	Desktop computers/laptops	For both trainer's and trainee' use	25 pcs	1:1
C	Consumable materials			
7.	Assorted colour of whiteboard markers		10 pcs	10:1
8.	Assorted timber		Sufficient	
9.	Assorted nails		Sufficient	
10.	Pegs		Sufficient	
11.	Mortar		Sufficient	

12.	Quarry dust		Sufficient	
13.	Cement/ lime		Sufficient	
14.	Clay		Sufficient	
15.	Grease/oil		Sufficient	
D	Tools and Equipment			
16.	Technical Drawing Instruments (Rulers, Protractors, Compasses, Set-squares)		25 Sets	1:1
17.	Tape measure		5 pcs	1:5
18.	Mason Hammer		5 pcs	1:5
19.	Shovel		5 pcs	1:5
20.	Bucket		5 pcs	1:5
21.	Wheelbarrow		5 pcs	1:5
22.	Trowel		5 pcs	1:5
23.	Wood Float		5 pcs	1:5
24.	Steel float		5 pcs	1:5
25.	Spade		5 pcs	1:5
26.	Moulds		5 pcs	1:5

MODULE II

BUILDING SITE PRELIMINARY WORKS I

UNIT CODE: 0732 351 04A

TVET CDACC UNIT CODE: CON/CU/BLD/CR/01/4/MA

UNIT DURATION: 120 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Learning: perform building site preliminary works

Unit Description

This unit describes the competence required to perform building site preliminary works.

It involves carrying out site clearance, carrying out site hoarding and building construction site amenities.

Summary of Learning Outcomes:

By the end of this unit, the trainee should be able to;

S/No	Learning Outcomes	Durations (hours)
1.	Carry out site clearance	30
2.	Carry out site hoarding	40
3.	Build construction site amenities	50
	Total	120

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Carry out site clearance	1.1 PPEs 1.1.1 Types 1.1.2 Uses 1.2 Site clearance tools and equipment 1.2.1 Slasher 1.2.2 Mattock	• Practical • Projects • Third Party Reports • Written tests

	1.2.3 Rake 1.2.4 Sledge hammer 1.2.5 Fork Jembe 1.2.6 Machete 1.2.7 Uses 1.3 Site clearance 1.3.1 Demarcation 1.3.2 Demolition 1.3.3 Grabbing of shrubs 1.3.4 Diversion of water channels and services 1.3.5 Site stripping 1.3.6 Trim hedges 1.4 Methods of site clearance 1.4.1 Mechanical methods 1.4.2 Manual methods Explosives /Detonators 1.5 Demolition 1.6 Methods of debris disposal 1.6.1 Incineration 1.6.2 Composting 1.6.3 Recycling	
2. Carry out site hoarding	2.1 PPEs 2.1.1 Types 2.1.2 Uses 2.2 Hoarding tools 2.2.1 Types 2.2.2 Uses 2.3 Hoarding materials 2.3.1 GI sheets 2.3.2 Assorted Timber 2.4 Demarcation and Erection of hoarding 2.5 Hoarding erected as per work	• Practical • Projects • Third Party Reports • Written tests

	requirements 2.6 Dismantling of hoarding 2.7 Storage of hoarding materials	
3. Build construction site amenities	3.1 PPEs 3.1.1 Types 3.1.2 Uses 3.2 Tools and equipment 3.2.1 Types 3.2.2 Uses 3.3 Site amenities materials assembling 3.3.1 Types 3.3.2 Construction 3.4 Site amenities are constructed as per the site plan 3.5 Demolition of Site amenities 3.6 Storage of site amenity materials	• Projects • Written tests • Practical tests • Oral tests

Suggested Methods of Instruction

- Practical
- Project
- Demonstrations
- Group Discussions
- Direct instructions

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Reference materials i.e. textbooks	For trainee's use	14 pcs	1:2

2.	Rolls Flip Charts	For both trainer's and trainee' use	5 pcs	1:5
B	Learning Facilities & infrastructure			
3.	Whiteboards	For trainer's use	1 pc	1:25
4.	Workshop/ construction site	10m by 10m	1	4:1
5.	Lecture room	9m by 8m	1	
6.	Desktop computers/laptops	For both trainer's and trainee' use	25 pcs	1:1
C	Consumable materials			
5.	Assorted colour of whiteboard markers	For trainer's use	10 pcs	10:1
6.	Assorted timber	For trainee's use	Sufficient	
7.	Assorted nails	For trainee's use	Sufficient	
8.	Pegs	For trainee's use	Sufficient	
9.	Hoarding materials	For trainee's use	Sufficient	
10.	Chalk lime	For trainee's use	Sufficient	
D	Tools and Equipment			
11.	Spirit Level	For trainee's use	5 pcs	1:5
12.	1. Builders Line	For trainee's use	5 pcs	1:5
13.	2. Mason Square	For trainee's use	5 pcs	1:5
14.	3. Mason Hammer	For trainee's use	5 pcs	1:5
15.	4. Claw Hammer	For trainee's use	5 pcs	1:5
16.	5. Shovel	For trainee's use	5 pcs	1:5
17.	6. Bucket	For trainee's use	5 pcs	1:5
18.	7. Wheelbarrow	For trainee's use	5 pcs	1:5
19.	8. Claw bar	For trainee's use	5 pcs	1:5
20.	9. Panga	For trainee's use	5 pcs	1:5
21.	10. Handsaw	For trainee's use	5 pcs	1:5

BUILDING TEMPORARY WORKS

UNIT CODE: 0732 453 05A

TVET CDACC UNIT CODE: CON/CU/BLD/CR /02/4/MA

UNIT DURATION: 120 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform Building Temporary Works.

Unit Description

This unit specifies the competencies required to perform building temporary works efficiently and safely. It involves performing timbering works, preparing building formwork, preparing construction scaffolds and preparing building shores.

Summary of Learning Outcomes

S/No	Learning Outcomes	Durations (Hours)
1.	Perform timbering works	30
2.	Prepare building formwork	30
3.	Prepare construction scaffolds	30
4.	Prepare building shores	30
	Total	120

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Perform timbering works	1.1 PPEs 1.1.1. Types 1.1.2. Uses 1.1.3. Identification 1.2 Timbering tools and equipment 1.2.1. Types 1.2.2. Uses 1.3 Trench timbering 1.4 Erection of timbering 1.5 Dismantling of timbering 1.6 Storage of timbering materials 1.6.1 Methods	• Practical • Projects • POE evaluation • Third party reports • Written tests

2. Prepare building formwork	2.1 PPEs 2.1.1. Types 2.1.2. Uses 2.2 Building formwork tools and equipment 2.2.1. Types 2.2.2. Uses 2.3 Preparation of formwork 2.3.1. Shapes and sizes 2.4 Erection of formwork 2.5 Striking of formwork 2.6 Storage of formwork materials 2.6.1. Methods	• Practical • Projects • POE evaluation • Third party reports • Written tests
3. Prepare construction scaffolds	3.1 PPEs 3.1.1. Types 3.1.2. Uses 3.2 scaffold tools and equipment 3.2.1. Types 3.2.2. Uses 3.3 Scaffold materials 3.3.1. Types 3.3.2. Uses 3.4 Erection of scaffold 3.4.1. Types 3.5 Dismantling of scaffold 3.6 Storage of scaffold materials 3.6.1. Methods	• Practical • Projects • POE evaluation • Third party reports • Written tests
4. Prepare building shores	4.1 PPEs 4.1.1. Types 4.1.2. Uses 4.2 Shoring tools and equipment 4.2.1. Types 4.2.2. Uses 4.3 Shoring materials 4.3.1. Types 4.3.2. Preparation 4.3.3. Shapes and sizes 4.4 Building shores size is prepared as per work requirement 4.5 Erection of shores	• Practical • Projects • POE evaluation • Third party reports • Written tests

	4.6 Dismantling of shores 4.7 Storage of shoring materials 13.	
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Suggested Methods of Instruction

- Practical
- Projects
- Demonstrations
- Group Discussion
- Direct Instruction

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Desktop computers/laptops	Anny model	5	1:5
2.	Tape measures	5 metres	25	1:1
3.	Builders line		5	1:5
4.	Claw hammer		5	1:5
5.	Mason square		5	1:5
6.	Timber	Assorted	Sufficient	
7.	Nails	Assorted	Sufficient	
8.	Claw bar		5	1:5
9.	Panga		5	1:5
10.	Jembe		5	1:5
11.	Pick axe		5	1:5
12.	Iron sheets	Assorted	Sufficient	
13.	Hand saw		5	1:5
14.	Mallet		5	1:5
B	Learning Facilities & Infrastructure			
15.	Lecture Room	(9* 8 sq. metres)	1	1:25
16.	Workshop/		1	1:25

	construction site			
17.	Internet Connection	System	1	1:25
18.	Whiteboard	Glass, melamine, porcelain	1	1:25

MODULE III

DIGITAL LITERACY

UNIT CODE: 0611 451 06A

TVET CDACC UNIT CODE: CON/CU/BLD/BC/01/5/MA

UNIT DURATION: 40 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: **Apply Digital Literacy**

Unit Description

This unit covers the competencies required to demonstrate digital literacy. It involves operating computer devices, solving tasks using the Office suite, managing data and information, performing online communication and collaboration, applying cybersecurity skills and job entry techniques, and performing jobs online.

Summary of Learning Outcomes

Learning Outcomes	DURATION (HOURS)
1. Operate Computer Devices	6
2. Solve Tasks Using Office Suite	14
3. Manage Data and Information	6
4. Perform Online Communication and Collaboration	4
5. Apply Cyber security Skills	4
6. Perform Online Jobs	4
7. Apply job entry techniques.	2
TOTAL	40

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Operate computer devices	1.1. Meaning and importance of digital literacy 1.2. Functions and Uses of Computers 1.3. Classification of computers 1.4. Components of a computer system 1.5. Computer Hardware 1.5.1. The System Unit E.g. Motherboard, CPU, casing 1.5.2. Input Devices e.g. Pointing, keying, scanning, voice/speech recognition, direct data capture devices. 1.5.3. Output Devices e.g. hardcopy output and softcopy output 1.5.4. Storage Devices e.g. main memory e.g. RAM, secondary storage (Solid state devices, Hard Drives, CDs & DVDs, Memory cards, Flash drives 1.5.5. Computer Ports e.g. HDMI, DVI, VGA, USB type C etc. 1.6. Classification of computer software 1.7. Operating system functions 1.8. Procedure for turning/off a computer 1.9. Mouse use techniques 1.10. Keyboard Parts and Use Techniques 1.11. Desktop Customization 1.12. File and Files Management using an operating system 1.13. Computer Internet Connection Options 1.13.1. Mobile Networks/Data Plans 1.13.2. Wireless Hotspots 1.13.3. Cabled (Ethernet/Fiber) 1.13.4. Dial-Up	• Observation • Written assessment • Oral assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	1.13.5. Satellite 1.14. Computer external devices management 1.14.1. Device connections 1.14.2. Device controls (volume controls and display properties)	
2. Solve tasks using Office suite	2.1. Meaning and Importance of Word Processing 2.2. Examples of Word Processors 2.3. Working with word documents 2.3.1. Open and close word processor 2.3.2. Create a new document 2.3.3. Save a document 2.3.4. Switch between open documents 2.4. Enhancing productivity 2.4.1. Set basic options/preferences 2.4.2. Help resources 2.4.3. Use magnification/zoom tools 2.4.4. Display, hide built-in tool bar 2.4.5. Using navigation tools 2.5. Typing Text 2.6. Document editing (copy, cut, paste commands, spelling and Grammar check) 2.7. Document formatting 2.7.1. Formatting text 2.7.2. Formatting paragraph 2.7.3. Formatting styles 2.7.4. Alignment	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	2.7.5. Creating tables 2.7.6. Formatting tables 2.8. Graphical objects 2.8.1. Insert object (picture, drawn object) 2.8.2. Select an object 2.8.3. Edit an object 2.8.4. Format an object 2.9. Document Print setup 2.9.1. Page layout, 2.9.2. Margins set up 2.9.3. Orientation. 2.10. Word Document Printing 2.11. Meaning & Importance of electronic spreadsheets 2.12. Components of Spreadsheets 2.13. Application areas of spreadsheets 2.14. Using spreadsheet application. 2.14.1. Parts of Excel screen: ribbon, formula bar, active cell, name box, column letter, row number, Quick Access Toolbar. 2.14.2. Cell Data Types 2.14.3. Block operations 2.14.4. Arithmetic operators (formula bar (-, +, *, /). 2.14.5. Cell Referencing 2.15. Data Manipulation 2.15.1. Using Functions (Sum, Average, Sum IF, Count, Max, Max, IF,	

Learning Outcome	Content	Suggested Assessment Methods
	Rank, Product, mode etc.) 2.15.2. Using Formulae 2.15.3. Sorting data 2.15.4. Filtering data 2.15.5. Visual representation using charts 2.16. Worksheet printing 2.17. Electronic Presentations 2.18. Meaning and Importance of electronic presentations 2.19. Examples of Presentation Software 2.20. Using the electronic presentation application 2.20.1. Parts of the PowerPoint screen (slide navigation pane, slide pane, notes, the ribbon, quick access toolbar, and scroll bars). 2.20.2. Open and close presentations 2.20.3. Creating Slides (Insert new slides, duplicate, or reuse slides.) 2.20.4. Text Management (insert, delete, copy, cut and paste, drag and drop, format, and use spell check). 2.20.5. Use magnification/zoom tools 2.20.6. Apply or change a theme. 2.20.7. Save a presentation 2.20.8. Switch between open presentations 2.21. Developing a presentation 2.21.1. Presentation views	

Learning Outcome	Content	Suggested Assessment Methods
	2.21.2. Slides 2.21.3. Master slide 2.22. Text 2.22.1. Editing text 2.22.2. Formatting 2.22.3. Tables 2.23. Charts 2.23.1. Using charts 2.23.2. Organization charts 2.24. Graphical objects 2.24.1. Insert, manipulate 2.24.2. Drawings 2.25. Prepare outputs Applying slide effects and transitions 2.26. Check and deliver 2.26.1. Spell check a presentation 2.26.2. Slide orientation 2.26.3. Slide shows, navigation 2.27. Print presentations (slides and handouts)	
3. Manage Data and Information	3.1. Meaning of Data and information 3.2. Importance and Uses of data and information 3.3. Types of internet services 3.3.1. Communication Services 3.3.2. Information Retrieval Services 3.3.3. File Transfer 3.3.4. World Wide Web Services 3.3.5. Web Services 3.3.6. Automatic Network Address Configuration	• Observation • Portfolio of Evidence • Practical assessment • Project • Written assessment • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	3.3.7. Newsgroup 3.3.8. Ecommerce 3.4. Types of Internet Access Applications 3.5. Web browsing concepts Key concepts Security and safety 3.6. Web browsing 3.6.1. Using the web browser 3.6.2. Tools and settings 3.6.3. Clearing Cache and cookies 3.6.4. URIs 3.6.5. Bookmarks 3.6.6. Web outputs 3.7. Web based information 3.7.1. Search 3.7.2. Critical evaluation of information 3.7.3. Copyright, data protection 3.8. Downloads Management 3.9. Performing Digital Data Backup (Online and Offline) 3.10. Emerging issues in internet	
4. Perform online communication and collaboration	4.1. Netiquette principles 4.2. Communication concepts 4.2.1. Online communities 4.2.2. Communication tools 4.2.3. Email concepts 4.3. Using email 4.4.1. Sending email 4.4.2. Receiving email	• Observation • Portfolio of Evidence • Project • Written assessment • Practical

Learning Outcome	Content	Suggested Assessment Methods
	4.4.3. Tools and settings 4.4.4. Organizing email 4.5. Digital content copyright and licenses 4.6. Online collaboration tools 4.6.1. Online Storage (Google Drive) 4.6.2. Online productivity applications (Google Docs & Forms) 4.6.3. Online meetings (Google Meet/Zoom) 4.6.4. Online learning environments 4.6.5. Online calendars (Google Calendars) 4.6.6. Social networks (Facebook/Twitter - Settings & Privacy) 4.7. Preparation for online collaboration Common setup features Setup 4.8. Mobile collaboration 4.8.1 Key concepts 4.8.2 Using mobile devices 4.8.3 Applications 4.8.4 Synchronization	assessment • Oral assessment
5. Apply cybersecurity skills	5.1. Data protection and privacy 5.1.1. Confidentiality of data/information 5.1.2. Integrity of data/information 5.1.3. Availability of data/information 5.2. Internet security threats 5.2.1. Malware attacks	• Observation • Portfolio of Evidence • Project • Written assessment • Practical

Learning Outcome	Content	Suggested Assessment Methods
	5.2.2. Social engineering attacks 5.2.3. Distributed denial of service (DDoS) 5.2.4. Man-in-the-middle attack (MitM) 5.2.5. Password attacks 5.2.6. IoT Attacks 5.2.7. Phishing Attacks 5.2.8. Ransomware 5.2. Computer threats and crimes 5.3. Cybersecurity control measures 5.3.1. Physical Controls 5.3.2. Technical/Logical Controls (Passwords, Pins, Biometrics) 5.3.3. Operational Controls 5.4. Laws governing protection of ICT in Kenya 5.4.1. The Computer Misuse and Cybercrimes Act No. 5 of 2018 5.4.2. The Data Protection Act No. 24 Of 2019	assessment • Oral assessment
6. Perform Online Jobs	6.1. Introduction to online working 6.2. Types of online Jobs 6.3. Online job platforms 6.3.1. Remo task 6.3.2. Data annotation tech 6.3.3. Cloud worker 6.3.4. Upwork 6.3.5. Oneforma 6.3.6. Appen	• Observation • Portfolio of Evidence • Project • Written assessment • Practical assessment

Learning Outcome	Content	Suggested Assessment Methods
	6.4. Online account and profile management 6.5. Identifying online jobs/job bidding 6.6. Online digital identity 6.7. Executing online tasks Management of online payment accounts.	• Oral assessment
7. Apply job entry techniques	7.1. Types of job opportunities 7.1.1. Self-employment 7.1.2. Service provision 7.1.3. product development 7.1.4. salaried employment 7.2. Sources of job opportunities 7.3. Resume/ curriculum vitae 7.3.1. What is a CV 7.3.2. How long should a CV be 7.3.3. What to include in an AC 7.3.4. Format of CV 7.3.5. How to write a good CV 7.3.6. Don'ts of writing a CV 7.4. Job application letter 7.4.1. What to include 7.4.2. Addressing a cover letter 7.4.3. Signing off a cover letter 7.5. Portfolio of Evidence 7.5.1. Academic credentials 7.5.2. Letters of commendations 7.5.3. Certification of participations 7.5.4. Awards and decorations 7.6. Interview skills 7.6.1. Listening skills 7.6.2. Grooming	• Observation • Oral assessment • Portfolio of evidence • Third party report • Written assessment

Learning Outcome	Content	Suggested Assessment Methods
	7.6.3. Language command 7.6.4. Articulation of issues 7.6.5. Body language 7.6.6. Time management 7.6.7. Honesty 7.7. Generally knowledgeable in current affairs and technical area	

Suggested Methods Instruction

- Instructor-led facilitation using active learning strategies
- Demonstration by trainer
- Practical work by trainees
- Viewing of related videos
- Group discussions
- Project
- Role play
- Case study

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	Recommended publisher	8 pcs	1:3
2.	Samples of CVs	Various formats	5	1:5
3.	Internet connection	Reliable	-	-
4.	White board	For trainer's use	1	-

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
B	Learning Facilities & infrastructure			
5.	Lecture/theory room	72 Square Meter	1	1:25
6.	Computer Lab	96 Square Meter	1	1:25
C	Consumable materials			
7.	Ink	Assorted Colours for trainer's use	500ml per term	-
8.	White board Marker	Refillable type	10 pcs per term	-
9.	Printing papers	sufficient	-	-
D	Tools and Equipment			
10.	Computers	Latest version with: Windows/Linux/Macintosh Operating System, Microsoft Office Software, Google Workspace Account, Antivirus Software	25	1:1
11.	Projector	Latest version	1	1:25
12.	External storage media	Latest version	25	1:1
13.	Laptop	Intel core i5	25	1:1
14.	printers	Latest version	2	1:13

COMMUNICATION SKILLS

UNIT CODE: 0031 441 07A

TVET CDACC UNIT CODE: CON/CU/BLD/BC/02/5/MA

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Communication Skills

Duration of Unit: 40 hours

Unit Description

This unit covers the competencies required to apply communication skills. It involves applying communication channels, written communication skills, non-verbal skills, oral, and group communication skills.

Summary of Learning Outcomes

S/NO	Learning Outcomes	Duration (Hours)
1.	Apply Communication Channels.	4
2.	Apply Written Communication Skills.	10
3.	Apply Non-Verbal Skills.	6
4.	Apply Oral Communication Skills.	8
5.	Apply Group Communication Skills.	12
	TOTAL	40

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply communication channels	1.1 Communication process 1.2 Principles of effective communication 1.3 Channels medium modes of communication 1.4 Factors to consider when selecting a channel of communication 1.5 Barriers to effective communication 1.6 Flow patterns of communication 1.7 Sources of information 1.8 Organizational policies	• Oral questions • Written assessment • Observation • Portfolio of Evidence • Practical assessment • Third party report
2. Apply written communication skills	2.1 Types of written communication 2.2 Elements of communication 2.3 Organization requirements for written communication	• Oral assessment • Written assessment • Observation • Portfolio of Evidence • Practical assessment • Third party report
3. Apply non-verbal communication skills	3.1 Utilize body language and gestures 3.2 Apply body posture 3.4 Apply workplace dressing code	• Oral assessment • Written assessment • Observation • Portfolio of Evidence • Practical assessment • Third party report
4. Apply oral communication skills	4.1 Types of oral communication pathways 4.2 Effective questioning	• Oral assessment • Written assessment • Observation

Learning Outcome	Content	Suggested Assessment Methods
	techniques 4.3 Workplace etiquette 4.4 Active listening	• Portfolio of Evidence • Practical assessment • Third party report
5. Apply group discussion skills	5.1 Establishing rapport 5.2 Facilitating resolution of issues 5.3 Developing action plans 5.4 Group organization techniques 5.5 Turn-taking techniques 5.6 Conflict resolution techniques 5.7 Team-work	• Oral assessemnt • Written assessment • Observation • Portfolio of Evidence • Practical assessment

Suggested Methods of Instruction

- Discussion
- Roleplaying
- Simulation
- Direct instruction
- Demonstration
- Field trips
- Viewing of related videos

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
	Charts	• Flip Charts • Rules and	5	1:5

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
		Regulations		
	Report Writing Templates	Printed copies and softcopies	25	1:1
	Assorted Markers	whiteboard markers and permanent	50	1:1
	Samples Of CVS	Printed copies and softcopies	5	1:5
	External Storage Media	Flash disks, Compass Disks; Re- Writable	1	1:25
	Smartboard/Smart TV (Where Applicable)	LCD or projector	20	1:25
B	Learning Facilities & Infrastructure			
7	Lecture/Theory Room	(9* 8 sq. metres)	1	1:25
8	Workshop	(10* 15 sq. metres)	1	1:25
9	Internet Connection	WI-FI, Dial-Up, Cable, Fixed- wireless,	1	1:25
C	Consumable Materials			
1	Flashcards	Alphabet, Numbers, Math	25	1:1

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
1.	Printing Papers	Sizes A4, A3, A2 etc	5 reams	1:5
D	Tools And Equipment			
1.	Computers/Laptops	Any model	1	1:25
2.	Projector	LED.LCD, Laser	5	1:5
3.	Printer	Inkjet, LaserJet	1	:25
4.	Computers Software:	•Windows/Linux/Ma cintosh Operating System •Microsoft Office Software •Google Workspace Account - Antivirus Software	1	1:1
5.	Whiteboard	Glass, melamine, porcelain	1	1:25
6.	Mobile Phones	Smartphones	5	1:5

BASIC MATHEMATICS I

UNIT CODE: 0541 451 08A

TVET CDACC UNIT CODE: CON/CU/BLD/CC/01/5/MA

Relationship to Occupational Standards

This unit addresses the unit of competency: Apply Basic Mathematics I

Duration of Unit: 60 Hours

UNIT DESCRIPTION:

This unit describes the competencies required in applying basic mathematics. It involves carrying out mensuration, applying basic arithmetic, applying basic algebra and applying statistics.

Summary of Learning Outcomes

S/NO	LEARNING OUTCOMES	DURATION(HOURS)
1.	Carry out basic Mensuration	10
2.	Apply basic arithmetic	10
3.	Apply basic Algebra	20
4.	Apply basic Statistics	20
	TOTAL	60

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Carry out basic Mensuration	1.1 Units of measurement 1.2 Conversion of units 1.3 Perimeters and areas 1.4 Volume and Surface area of solids 1.5 Area of irregular figures	• Written assessments • Assignments • Supervised exercises
2. Apply basic arithmetic	2.1 Whole numbers 2.2 Simple fractions 2.3 Decimals 2.4 Percentage	• Written assessments • Assignments • Supervised exercises

	2.5 Place value 2.6 Ranges 2.7 Rounding off 2.8 Arithmetic Percentages and Proportions 2.9 Decimals 2.10 Standard forms	
3. Apply basic Algebra	3.1 Indices and logarithms 3.2 Linear equations 3.3 Use of a calculator to solving problems 3.4 Simultaneous equations 3.5 Simple algebraic equations 3.6 Form of simple algebraic equations	• Written assessments • Assignments • Supervised exercises
4. Apply basic Statistics	4.1 Identification of grouped and ungrouped data 4.2 Ungrouped data organization 4.3 Frequency table data representation 4.4 Mean, Mode and Median of grouped and ungrouped data 4.5 Charts 4.6 Chart data interpretation.	• Written assessments • Assignments • Supervised exercises

Suggested Delivery Methods

- Practical
- Projects
- Group discussion
- Direct instruction

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Scientific Calculators	Casio FX-82MS	25	1:1
2.	Rulers	30 CM,15 CM	25	1:1
3.	Pencils	HB, H,2H	25	1:1
4.	Erasers	Staedtler	25	1:1
5.	Charts	with presentations of data	25	1:1
6.	Graph books	A4, A3	25	1:1

TECHNICAL DRAWING

UNIT CODE: 0732 451 09A

TVET CDACC UNIT CODE: CON/CU/BLD/CC/02/5/MA

Relationship to Occupational Standards

This unit addresses the unit of competency: **Apply Technical Drawing**

Duration of Unit: 80 Hours

UNIT DESCRIPTION

This unit covers the competencies required to prepare and interpret technical drawings. It involves competencies to use and maintain drawing equipment and materials. It also involves producing plain geometry drawings, solid geometry drawings, pictorial and orthographic drawings

Summary of Learning Outcomes

S/NO	LEARNING OUTCOME	DURATION(HOURS)
1.	Use and maintain drawing equipment and materials	10
2.	Produce plane geometry drawings	20
3.	Produce solid geometry drawings	25
4.	Produce pictorial and orthographic drawings	25
5.	Interpret building drawings	20
	TOTAL	100

Learning Outcomes, Content and Suggested Assessment Methods:

Learning Outcome	Content	Suggested Assessment Methods
1. Select, use and maintain drawing equipment and materials	1.1 Identification of drawing equipment 1.1.1. Drawing boards 1.1.2. T and set squares 1.1.3. drawing sets 1.2 Identification of drawing	• Practical • Written tests • Third party reports • Oral assesment

Learning Outcome	Content	Suggested Assessment Methods
	materials 1.2.1. Drawing papers 1.2.2. Pencils 1.2.3. Erasers 1.2.4. masking tapes 1.2.5. paper clips 1.3 Use and maintenance of drawing equipment's 1.4 Use of drawing materials 1.5 Waste material disposal- EMCA 1999	
2. Produce plane geometry drawings	2.1 Types of lines used in drawings 2.2 Construction of geometric forms 2.2.1. Circles 2.2.2. Triangles 2.2.3. Rectangles 2.2.4. Parallelogram 2.2.5. Polygons 2.2.6. Pyramids 2.2.7. conic sections 2.2.8. prisms, loci 2.3 Construction of angles 2.3.1. -types 2.4 Measurement of different angles 2.5 Bisection of different angles 2.6 Freehand sketching of geometric forms, tools and equipment	• Practical • Written tests • Third party reports • Oral assesment

Learning Outcome	Content	Suggested Assessment Methods
3. Produce solid geometry drawings	3.1 Interpretation of sketches and drawings of patterns 3.2 Development of truncated and untruncated regular solids	• Practical • Written tests • Third party reports • Oral assesment
4. Produce orthographic and pictorial drawings.	4.1 Meaning of abbreviations and symbols 4.2 Drawing and interpretation of orthographic drawings 4.2.1. 1st angle 4.2.2. 3rd angle 4.3 Dimensioning of orthographic drawings 4.4 Interpretation and production of isometric drawings 4.5 interpretation of oblique drawings	• Practical • Written tests • Third party reports • Oral assessment
5. Interpret building drawings.	5.1 Symbols and abbreviations 5.1.1. -Identification 5.1.2. -Meaning of symbols 5.2 interpretation of Dimensions and annotations 5.3 Interpretation of building section elements 5.4 Interpretation of building elevations.	• Practical • Written tests • Third party reports • Oral assessment

Suggested Methods of Delivery Dimensions and annotations Demonstration by trainer

- Practice by the trainee
- Group discussions

- Direct instruction

Recommended Resources for 25 trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1	T-square	800mm-plastic or wooden	25	1:25
2	Set squares	30,60 degrees, 45 degrees by 300mm	25	1:25
3	Drawing sets	Helix technical drawing set	25	1:25
4	French curves	Hyperbola, parabola and elliptical (Clear Tint)	25	1:25
5	Drawing tables	Dimensions 600x800 wooden or plastic	25	1:25
6.	Building drawings	Architectural drawing Structural drawing Mechanical drawing Electrical drawing	5	1:5
B	Learning Facilities & infrastructure			
7.	Lecture/theory room	(9* 8 sq. metres)	1	1:25
8.	Drawing room	15* 10 sq. metres)	1	1:25

BUILDING SUBSTRUCTURE WORKS

UNIT CODE: 0732 451 10A

TVET CDACC UNIT CODE: CON/CU/BLD/CR/01/5/MA

UNIT DURATION: 120 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform Building Substructure Works

Unit Description

This unit specifies the competencies required to performing building substructure works. It involves, executing foundation excavation works, laying foundation concrete, carrying out foundation walling and laying out construction oversite concrete.

Summary of Learning Outcomes

S/NO	LEARNING OUTCOME	DURATION(HOURS)
1.	Execute foundation excavation works	20
2.	Lay foundation concrete	40
3.	Carry out foundation walling	30
4.	Lay out construction over-site concrete	30
	TOTAL	120

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1.Execute foundation excavation works	1.1 PPEs 1.1.1 Types 1.1.2 Uses 1.2 Types of foundation 1.3 Trench excavation tools and equipment 1.3.1 Types 1.3.2 Uses 1.4 Trench timbering materials	•Practical •Projects •POE evaluation •Third party reports •Written tests

	1.4.1 Preparation 1.5 Trench excavation 1.6 Erection of trench timbering 1.7 Trench Dewatering 1.7.1 Methods 1.8 Anti – termite treatment 1.8.1 Uses	
2. Lay foundation concrete	2.1 PPEs 2.2 Concrete works tools and equipment identification 2.3 Concrete class 2.1.1 Class 15 2.1.2 Class 20 2.1.3 Class 25 2.1.4 Class 30 2.4 Batching of Concrete materials 2.4.1 Cement 2.4.2 Fine aggregates 2.4.3 Course aggregates 2.4.4 Water 2.5 Fixing of Reinforcement bars 2.6 Mixing of Batched materials 2.7 Placing of Fresh concrete 2.8 Compaction of concrete 2.9 Curing	
3 Carry out foundation walling	3.1 PPEs 3.2 Foundation wall construction tools and equipment	

	3.2.1 Trowel 3.2.2 Spade 3.2.3 Wheelbarrows 3.2.4 Building line 3.2.5 Plumb bob 3.2.6 Spirit level 3.2.7 Mason square 3.3 Preparation of <i>Foundation wall construction materials</i> 3.4 Transformation of Foundation wall dimensions. 3.5 Foundation wall is erection 3.6 Curing process 3.7 Backfilling	
4 lay out construction over-site concrete	4.1 PPEs 4.2 Tools and equipment 4.3 Ground construction materials 4.3.1 Hard-core 4.3.2 Blinding 4.3.3 Damp-proof membrane 4.3.4 BRC 4.4 Functional requirements of oversite concrete 4.5 Laying of hard-core 4.5.1 Functions of hard-core 4.5.2 Procedure 4.6 Formwork erection 4.7 Blinding 4.7.1 Materials	• Practical assessment • Projects • Third party report • Written assessment • Oral assessment • Portfolio of evidence

	4.7.2 Function of blinding. 4.7.3 Procedure 4.8 Termite control 4.8.1 Importance 4.9 Laying of DPM 4.9.1 Materials 4.9.2 Procedure 4.10 Laying of BRC mesh 4.10.1 Procedure 4.11 Over-site concrete construction materials 4.11.1 Cement 4.11.2 Fine aggregates 4.11.3 Course aggregates 4.11.4 Water 4.12 Concrete slab casting 4.13 Curing	
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Suggested Methods of Instruction

- Practical
- Projects
- Demonstrations
- Group Discussion
- Direct Instruction

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Desktop computers/laptops	Anny model	5	1:5

2.	Tape measures	5 metres	25	1:1
3.	Builders line		5	1:5
4.	Mason square		5	1:5
5.	Timber	Assorted	Sufficient	
6.	Nails	Assorted	Sufficient	
7.	Claw bar		5	1:5
8.	Panga		5	1:5
9. 5	Jembe		5	1:5
10.	Pick axe		5	1:5
11.	Cement/ lime		25 bags	1:1
12.	spade		5	1:5
13.	Trowel		25	1:1
14.	Spirit level		5	1:5
15.	Plumb bob		25	1:1
16.	Wood float		25	1:1
17.	Bar bender		5	1:5
18.	BRC Mesh/ Reinforcement bars		5	1:5
19.	Hardcore		10Ton	1:5
20.	Murram		10Ton	1:5
21.	Ballast		10Ton	1:5
22.	Sand		10Ton	1:5
23.	DPM		1roll	
24.	Anti- termite control		10L	1:5
25.	Concrete mixer		1	1:25
26.	Dumpy level		5	1:5
27.	Total Station		5	1:5
28.	Tamping rod/ poker vibrator		1	1:25
B	Learning Facilities & Infrastructure			

29.	Lecture Room	(9* 8 sq. metres)	1	1:25
30.	Workshop/ construction site		1	1:25
31.	Internet Connection	System	1	1:25
32.	Whiteboard	Glass, melamine, porcelain	1	1:25

DOORS AND WINDOWS

UNIT CODE: 0732451 11A

TVET CDACC UNIT CODE: CON/CU/BLD/CR/02/5/MA

UNIT DURATION: 50 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Install doors and windows

Unit Description

This unit describes the competencies required to install doors and windows. It involves Installing door and window frames, constructing and fixing door and windows and fixing iron mongeries.

Summary of Learning Outcomes

S/NO	LEARNING OUTCOME	DURATION(HOUR)
1.	To install door and window frames	20
2.	To construct and fix door and window shutters	20
3.	Fix iron mongeries	10
	TOTAL	50

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Install door and window frames	1.1 Preparation and interpretation of door and window schedule 1.2 Costing of doors and windows frames 1.3 Fixing door and window frames	• Written assessment • Practical • Projects • Oral
2. construct and fix door and window	2.1 Door and window <i>shutters</i> 2.1.1 Metallic	• Written assessment

shutters	2.1.2 Plastic 2.1.3 Glass 2.1.4 Wooden 2.2 Construction of Doors and window shutters 2.3 positioning of Doors and windows	• Practical • Projects • Oral
3. Fix iron mongeries	3.1 <i>Identification of Iron mongeries</i> 3.1.1 Fasteners 3.1.2 Peg Stay 3.1.3 Enclosures 3.1.4 Ball catchers 3.1.5 Locks 3.1.6 Hinges 3.1.7 Handles 3.1.8 Stoppers 3.1.9 Nails and screws 3.2 Fitting of Iron mongeries 3.3 Testing of Iron Mongeries	•

Suggested Methods of Instruction

- Viewing of related videos
- Discussion
- Direct Instruction
- Lecture
- Question and answer
- Site visits
- Practical

Recommended Resources for 20 Trainees

- Drawing room
- 10 Desktop computers/laptops
- 20 Drawing tables
- Drawing papers
- Internet connection
- 1 Projector
- 1 Whiteboard
- 1 roll Flip Chart
- Assorted colour of whiteboard markers
- 1 Blueprinter
- Drawing instruments
- Hand tools

- Carpentry workshop
- Tiles
- Lining boards
- Lime

EXTERNAL WORKS

UNIT CODE: 0732 451 12A

TVET CDACC UNIT CODE: CON/CU/BLD/CR/03/5/MA

UNIT DURATION: 60 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: execute external works

UNIT DESCRIPTION

This unit describes the competencies required to execute external works. It involves performing landscaping, constructing drainage system, laying external paving, and constructing gates and fences.

Summary of Learning Outcomes

S/No	Learning Outcomes	Durations (Hours)
1.	Perform landscaping	15
2.	Construct drainage system	15
3.	Lay external paving	15
4.	Construct gates and fences	15
	Total	60

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Perform landscaping	1.1 Determination of area for landscaping 1.2 Preparation of landscaping ground 1.3 Planting vegetative features 1.4 Method of irrigation 1.5 Beatification 1.5.1 Ornamental trees 1.5.2 Grassing 1.5.3 Flowers 1.5.4 Shrubs 1.5.5 Ground cover 1.5.6 Garden furniture 1.5.7 Garden lighting	• Written texts • Practical • Projects • Oral

2. Construct drainage system	2.1 Interpretation of Drainage drawings 2.2 Excavation <i>of Drainage channels</i> 2.2.1 Open channels 2.2.2 Closed channels 2.3 Backfilling, Levelling and compacting 2.4 <i>Laying of Drainage pipes</i> 2.4.1 concrete pipes 2.4.2 PVC pipes 2.4.3 GI pipes 2.4.4 PPR pipes 2.5 Construction of Drainage channels and collection chambers	• Written assessment • Practical • Projects • Oral
3. Lay external paving	3.1 Ground survey 3.2 Ground excavation 3.3 Base preparation 3.4 Levelling 3.5 Kerbs stones, surface drainage 3.6 Laying external paving 3.7 Marking	• Written assessment • Practical • Projects • Oral
4. Construct gates and fences	4.1 Gate location and orientation 4.2 Gate measurements 4.3 Gate supporting systems 4.3.1 RC columns 4.3.2 Stanchions 4.3.3 Stone Piers 4.4 Procurement and installation of the gate 4.5 Perimeter fence 4.6 Essential services 4.6.1 CCTVS cameras 4.6.2 security lighting 4.6.3 razor wire 4.6.4 Electric fence.	• Written assessment • Practical • Projects • Oral

Suggested Methods of Instruction

- Viewing of related videos
- Discussions
- Direct Instruction
- Lecture
- Question and answer
- Site visits
- Practicals

Recommended Resources for 20 Trainees

- 10 Desktop computers/laptops
- Internet connection
- 1 Projector
- 1 Whiteboard
- 1 roll Flip Chart
- Assorted colour of whiteboard markers
- Precast concrete kerbs
- Precast concrete paving units
- Quarry dust
- Sand
- Cement
- Ballast
- Construction tools
 - 10 Spade
 - 5 Wheelbarrows
 - 20 Painting brushes
 - 20 Trowels
 - 20 Spirit levels
 - 20 Screw drivers
 - 20 Hammers
 - 20 Plumb bobs
 - 20 Wooden float
 - 20 Steel float
 - 20 Head pans

MODULE IV

WORK ETHICS AND PRACTICES

UNIT CODE: 0417 441 13A

TVET CDACC UNIT CODE: CON/CU/BLD/BC/03/5/MA

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Apply Work Ethics and Practices.

Duration of Unit: 40 hours

Unit Description

This unit covers competencies required to demonstrate employability skills. It entails conducting self-management skills, promoting ethical work practices and values, promoting teamwork, maintaining professional and personal development, applying problem-solving skills and promoting customer care.

Summary of Learning Outcomes

S/NO	LEARNING OUTCOMES	DURATION(HOURS)
1.	Apply Self-Management Skills	10
2.	Promote Ethical Practices and Values	4
3.	Promote Teamwork	10
4.	Maintain Professional and Personal Development	10
5.	Apply Problem-Solving Skills	4
6.	Promote Customer Care.	2
	TOTAL	40

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply Self-Management Skills	1.1 Self-awareness 1.2 Formulating personal vision, mission, and goals	● Observation ● Written assessment ● Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	1.3 Healthy lifestyle practices 1.4 Strategies for overcoming work challenges 1.5 Emotional intelligence 1.6 Coping with Work Stress. 1.7 Assertiveness versus aggressiveness and passiveness 1.8 Developing and maintaining high self-esteem 1.9 Developing and maintaining positive self-image 1.10 Time management 1.11 Setting performance targets 1.12 Monitoring and evaluating performance targets	● Third party reports ● Portfolio of evidence ● Project ● Practical
2. Promote Ethical Work Practices and Values	2.1 Integrity 2.2 Core Values, ethics and beliefs 2.3 Patriotism 2.4 Professionalism 2.5 Organizational codes of conduct 2.6 Industry policies and procedures	● Observation ● Written assessment ● Oral assessment ● Third party reports ● Portfolio of evidence ● Project ● Practical
3. Promote Teamwork	3.1 Types of teams 3.2 Team building 3.3 Individual responsibilities in a team 3.4 Determination of team roles and objectives 3.5 Team parameters and relationships 3.6 Benefits of teamwork 3.7 Qualities of a team player 3.8 Leading a team 3.9 Team performance and evaluation 3.10 Conflicts and conflict resolution 3.11 Gender and diversity	● Observation ● Written assessment ● Oral assessment ● Third party reports ● Portfolio of evidence ● Project ● Practical

Learning Outcome	Content	Suggested Assessment Methods
	mainstreaming 3.12 Developing Healthy workplace relationships 3.13 Adaptability and flexibility 3.14 Coaching and mentoring skills	
4. Maintain Professional and Personal Development	4.1 Personal vs professional development and growth 4.2 Avenues for professional growth 4.3 Recognizing career advancement 4.4 Training and career opportunities 4.5 Assessing training needs 4.6 Mobilizing training resources 4.7 Licenses and certifications for professional growth and development 4.8 Pursuing personal and organizational goals 4.9 Managing work priorities and commitments 4.10 Dynamism and on-the-job learning	● Observation ● Written assessment ● Oral assessment ● Third party reports ● Portfolio of evidence ● Project ● Practical
5. Apply Problem-Solving Skills	5.1 Causes of problems 5.2 Methods of solving problems 5.3 Problem-solving process 5.4 Decision making 5.5 Creative thinking and critical thinking process in development of innovative and practical solutions	● Observation ● Written assessment ● Oral assessment ● Third party reports ● Portfolio of evidence ● Project ● Practical
6. Promote Customer Care	6.1 Identifying customer needs 6.2 Qualities of good customer service 6.3 Customer feedback methods 6.4 Resolving customer concerns 6.5 Customer outreach programs 6.6 Customer retention	● Observation ● Written assessment ● Oral assessment ● Third party reports ● Portfolio of evidence ● Project ● Practical

Suggested Methods of Instruction

- Instructor lead facilitation of theory using active learning strategies.
- Demonstrations
- Simulation/Role play
- Group Discussion
- Presentations
- Projects
- Case studies
- Assignments

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Charts	Flip Charts Rules and Regulations	5	1:5
2.	Markers	whiteboard markers and permanent markers	50	1:1
3.	Video clips Audio tapes	MP4, MP3	5	1:5
B	Learning Facilities & Infrastructure			
4.	Lecture/Theory Room	(9* 8 sq. metres)	1	1:25
5.	Internet Connection	WI-FI, Dial-Up, Cable, Fixed- wireless,	1	1:25
C	Consumable Materials			
6.	Flashcards	Alphabet, Numbers, Math	25	1:1
7.	Stationery	Printing Papers, and Exercise Books Sizes A4,	5 reams	1:5

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
		A3, A2 etc		
D	Tools And Equipment			
8.	Computers/Laptops	Any model	1	1:25
9.	Projector	LED.LCD, Laser	5	1:5
10.	Whiteboard	Glass, melamine, porcelain	1	1:25
11.	Mobile Phones	Smartphones	5	1:5

ENTREPRENEURIAL SKILLS

UNIT CODE: 0413 451 14A

TVET CDACC UNIT CODE: CON/OS/BLD/BC/04/5/MA

Relationship to occupational standards

This unit addresses the unit of competency: **Apply Entrepreneurial skills.**

Duration of unit: 40 hours

Unit Description:

This unit covers the competencies required to demonstrate an understanding of entrepreneurship. It involves demonstrating an understanding of financial literacy, applying entrepreneurial concepts, identifying entrepreneurship opportunities, applying business legal aspects, and developing business innovative strategies and business plans.

Summary of Learning Outcomes

S.NO	LEARNING OUTCOMES	DURATION(HRS)
1.	Apply financial literacy	6
2.	Apply the entrepreneurial concept	4
3.	Identify entrepreneurship opportunities	6
4.	Apply business legal aspects	6
5.	Innovate Business Strategies	6
6.	Develop a business plan	12
TOTAL		40

Learning Outcomes, Content, and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Apply financial literacy	1.1 Personal finance management 1.2 Balancing between needs and wants 1.3 Budget Preparation 1.4 Saving management 1.5 Factors to consider when deciding where to save 1.6 Debt management 1.7 Factors to consider before	• Observation • Project • Written assessment • Oral assessment • Third party report • Interviews

Learning Outcome	Content	Suggested Assessment Methods
	taking a loan 1.8 Investment decisions 1.9 Types of investments 1.10 Factors to consider when investing money 1.11 Insurance services 1.12 Insurance products available in the market 1.13 Insurable risks	
2. Apply entrepreneurial concept	2.1. Difference between Entrepreneurs and Business persons 2.2. Types of entrepreneurs 2.3. Ways of becoming an entrepreneur 2.4. Characteristics of Entrepreneurs 2.5. salaried employment and self-employment 2.6. Requirements for entry into self-employment 2.7. Roles of an Entrepreneur in an enterprise 2.8. Contributions of Entrepreneurship	• Observation • Project • Written assessment • Oral assessment • Third party report
3. Identify entrepreneurship opportunities	3.1. Sources of business ideas 3.2. Factors to consider when evaluating business opportunity 3.3. Business life cycle	• Observation • Project • Written assessment • Oral assessment • Third party report
4. Apply business legal aspects	4.1. Forms of business ownership 4.2. Business registration and licensing processing 4.3. Types of contracts and agreements 4.4. Employment laws 4.5. Taxation laws	• Observation • Project • Written assessment • Oral assessment • Third party report
5. Innovate business Strategies	5.1 Creativity in business 5.2 Innovative business strategies 5.3 Entrepreneurial Linkages 5.4 ICT in business growth and development	• Observation • Project • Written assessment • Oral assessment • Third party report
6. Develop Business Plan	6.1 Business description 6.2 Marketing plan 6.3 Organizational/Management	• Observation • Written assessment • Project

Learning Outcome	Content	Suggested Assessment Methods
	plan 6.4 Production/operation plan 6.5 Financial plan 6.6 Executive summary 6.7 Business plan presentation 6.8 Business idea incubation	• Oral assessment • Third party report

Suggested Delivery Methods

- Project (Business plan)
- Case studies
- Field trips
- Group Discussions
- Demonstration
- Question and answer
- Problem solving
- Experiential
- Team training
- Guest speakers

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Textbooks	The Lean Startup by Eric Ries.	10 pcs	1:2.5
		Zero to One by Peter Thiel	10 pcs	1:2.5
		Start with Why by Simon Sinek	10 pcs	1:2.5
B	Learning Facilities & infrastructure			
1.	Lecture/theory room	60m ²	1	1:25

2.	Computer workshop	160 m ²	1	1:25
3.	Computers	Operating System: 64-bit Windows 11 or 10 version 1809 or above Processor: 2.5 GHz (3+ GHz recommended), Memory: 8 GB (32GB recommended) Disk space: 10 GB Display: 1920 x 1080 resolution Display Card: 2 GB GPU (8 GB recommended) and DirectX 11 compliant (DirectX 12 recommended)	25 pcs	1:1
4.	Projector	high-resolution projectors with HDMI input	1	1:25
5.	Smartboard/Smart TV	Specifications: 77-inch interactive whiteboard with touch and pen functionality.	1	1:25
6.	Whiteboard	Traditional whiteboard	1	1:25
7.	Printers	With Print, Copy, Scan and Fax	2	1:12.5
C	Software			
8.	Operating systems	Windows/Linux/Macintosh Operating System	Installed in 25 computers	1:1
9.	Web Browsers	Chrome, Firefox, Edge, Safari	Installed in 25 computers	1:1
10.	Software for business planning	Live Plan, Bizplan	Installed in 25 computers	1:1
11.	Market research tools	Google Trends or SurveyMonkey or Statista	Installed in 25 computers	1:1
12.	Marketing platforms	Google Ads or Hootsuite	Installed in 25	1:1

			computers	
13.	Graphic design software	Adobe creative cloud or canvas	Installed in 25 computers	1:1
D	Consumables			
14.	Pens, pencils, rulers and paper	Whiteboard markers, 2H pencils, plastic rulers, A2 white papers	Enough	
15.	Printing papers	A4 and A3	Enough	

BASIC MATHEMATICS II

UNIT CODE: 0541 451 14A

TVET CDACC UNIT CODE: CON/CU/BLD/CC/03/5/MA

Relationship to Occupational Standards

This unit addresses the unit of competency: Apply Basic Mathematics II

Duration of Unit: 80 Hours

UNIT DESCRIPTION:

This unit describes the competencies required in applying basic mathematics. It involves performing trigonometry calculations, performing geometric calculations and applying linear graphs.

Summary of Learning Outcomes

S/NO	LEARNING OUTCOME	DURATION(HOURS)
1.	Perform Trigonometry calculations	30
2.	Perform geometric calculations	30
3.	Apply linear graphs	20
	TOTAL	80

Learning Outcomes, Content and Suggested Assessment Methods

1. Perform Trigonometry calculations	1.1 Trigonometric calculations 1.2 Trigonometric ratios 1.3 Trigonometric Calculations	• Written assessments • Assignments • Supervised exercises
2. Perform geometric calculations	2.1 Geometrical figures 2.2 Areas of figures 2.3 Apply Pythagoras' theorem	• Written tests • Assignments • Supervised exercises

3. Apply linear graphs	3.1 Data set identification 3.2 Scales in graphs 3.3 Axes labeling 3.4 Linear graph plotting 3.5 Distance against time 3.6 Temperature against time 3.7 Velocity against time 3.8 Graph analysis	• Written assessments • Assignments • Supervised exercises
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Suggested Delivery Methods

- Practical
- Projects
- Group discussion
- Direct instruction

Recommended Resources for 25 trainees

S/N o.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Scientific Calculators	Casio FX-82MS	25	1:1
2.	Rulers	30 CM,15 CM	25	1:1
3.	Pencils	HB, H,2H	25	1:1
4.	Erasers	Staedtler	25	1:1
5.	Charts	with presentations of data	25	1:1
6.	Graph books	A4, A3	25	1:1

CONSTRUCTION MATERIAL SCIENCE

UNIT CODE: 0722 451 15A

TVET CDACC UNIT CODE: CON/CU/BLD/CC/04/5/MA

Relationship to Occupational Standards

This unit addresses the unit of competency: Apply construction material science.

Duration of Unit: 60 Hours

UNIT DESCRIPTION

This unit describes the competence in applying Construction materials science. It involves identifying essential construction materials and their properties, selecting quality construction materials, testing construction material and handling and use of construction materials.

Summary of Learning Outcomes

S/NO	LEARNING OUTCOMES	DURATION(HOURS)
1.	Identify essential construction materials	10
2.	Identify properties of construction materials	10
3.	Select quality construction materials	10
4.	Test construction materials	10
5.	Handle construction materials	10
6.	Use construction materials	10
	TOTAL	60

Learning Outcomes, Content and Suggested Assessment Methods:

Learning Outcome	Content	Suggested Assessment Methods
1. Identify essential construction materials	1.1 Identification of construction material 1.1.1. walling materials 1.1.2. roofing materials 1.1.3. clay and clay products 1.1.4. glass and glass products 1.1.5. timber and timber products	• Practical • Written tests • Third party reports • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	1.1.6. reinforcing materials 1.1.7. finishing materials 1.1.8. roofing materials 1.1.9. finishing materials 1.1.10. binding materials 1.1.11. Glass and glass products 1.1.12. Bitumen and bituminous products	
2. Identify properties of construction materials	2.1 Identification of physical properties of construction materials 2.1.1. Porosity 2.1.2. Surface texture 2.1.3. Strength 2.1.4. Density 2.1.5. Thermal conductivity 2.1.6. Wear and tear 2.2 Identification of chemical properties of construction materials 2.2.1. Corrosion resistance 2.2.2. Chemical resistance 2.3 Identification of mechanical properties of construction materials 2.3.1. Toughness 2.3.2. Hardness 2.3.3. Fatigue 2.3.4. Stress and strain	• Practical • Written tests • Third party reports • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
	2.3.5. Creep and stress rapture 2.3.6. Strength	
3. Select quality construction materials	3.1 Analysis and evaluation of Cost implications of construction materials 3.2 Selection of Quality of construction materials 3.2.1. -cost 3.2.2. -availability 3.2.3. -requirement 3.3 Selection criteria of construction materials.	• Practical • Written tests • Third party reports • Oral assessment
4. Test construction materials	4.1 Sampling of construction materials 4.1.1. -random sampling 4.2 Test parameters 4.2.1. Compression 4.2.2. Weathering 4.2.3. Durability 4.2.4. Water absorption 4.2.5. Impurity tests 4.2.6. Tensile tests 4.2.7. Workability 4.2.8. Plasticity 4.2.9. Aggregates crushing value 4.2.10. Optimum moisture content 4.3 Testing of construction materials	• Practical • Written tests • Third party reports • Oral assessment

Learning Outcome	Content	Suggested Assessment Methods
5. Handle construction materials	5.1 Handling of identified construction materials 5.2 Identification of safety requirements of construction materials 5.3 Safety requirements of construction materials	• Practical • Written tests • Third party reports • Oral assessment
6. Use construction materials	6.1 Assembling of Construction materials, tools and equipment. 6.2 Preparation of construction materials 6.3 Requirement of construction materials 6.3.1. -uses	• Practical • Written tests • Third party reports • Oral assessment

Suggested Methods of Delivery

- Projects
- Group discussions
- Direct instruction
- Practical's
- Third party reports

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Charts	Procedures, rules and regulations, flow charts, Diagrams	5	1:5

2.	Stationery	A4 Exercise Pens Graph papers Eraser Pencils	25 each	1:1
B	Learning Facilities & infrastructure			
3.	Lecture/theory room	(9* 8 sq. metres)	1	1:25
4.	Equipped material testing lab	(15*10 sq. metres)	1	1:25

BUILDING SUPERSTRUCTURE WORKS

UNIT CODE: 0732 453 16A

TVET CDACC UNIT CODE: CON/CU/BLD/CR/04/5/MA

UNIT DURATION: 80 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Carry Out Building Super Structure Works

Unit Description

This unit specifies the competencies required to perform building super structure works. It involves: Erecting structural elements, laying of superstructure walls, carrying out upper floor construction and fixing building casements.

These entail activities that are carried out above the ground level of a building and also contribute greatly to the functionality and stability of the structure under construction.

Summary of Learning Outcomes

S/NO	LEARNING OUTCOMES	DURATION(HOURS)
1.	Erect structural elements	20
2.	Lay superstructure walls	20
3.	Carry out upper floor construction.	20
4.	Fix building casements	20
	TOTAL	80

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1.Erect structural	1.1 <i>PPEs</i> 1.2 Assembling of <i>Structural element</i>	• Practical • Projects

elements	<i>tools and equipment</i> 1.3 Preparation of <i>Structural element materials</i> 1.4 Fixing of Reinforcement bars 1.5 Erection of Formwork. 1.6 Placing of Concrete 1.7 Strucking of Formwork 1.8 Curing requirement	• POE evaluation • Third party reports • Written tests
2. Lay superstructure walls	2.1 PPEs 1.1.1. Types 1.1.2. Uses 2.2 Wall construction tools and equipment 1.2.1 Types 1.2.2 Uses 2.3 Wall construction 2.3.1 External 2.3.2 Partitioning 2.3.3 Cavity 2.4 Walling materials 2.4.1 Cement 2.4.2 Fine aggregate 2.4.3 Walling units 2.4.4 Hoop iron 2.4.5 Water 2.5 Screed 2.6 DPC 2.7 Erection of construction wall 2.8 Waste materials	• Practical • Projects • POE evaluation • Third party reports • Written tests

	2.8.1 Reuse 2.9 Superstructure wall opening 2.10 Superstructure wall curing	
3 Carry out upper floor construction	3.1 PPEs Types 3.1.1 Uses 3.2 Upper floors 3.2.1 Types Ribbed 3.2.2 Hollow pots 3.2.3 Flat slab 3.3 Upper floor construction tools and equipment 3.3.1 Types 3.3.2 Uses 3.3.3 Design considerations 3.4 Upper floor construction materials 3.5 Erection of formwork 3.6 Upper floor installation 3.7 Form work striking 3.8 Upper floor curing	• Practical • Projects • POE evaluation • Third party reports • Written tests
4 Fix building casements	4.1 PPEs 4.2 Building Casement positioning 4.3 Obtaining of Building Casements 4.4 Fixing of Building Casements	• Practical • Projects • POE evaluation • Third party reports • Written tests

Suggested Methods of Instruction

- Practical
- Projects
- Demonstrations
- Group Discussion
- Direct Instruction

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Desktop computers/laptops	Any model	5	1:5
2.	Tape measures	5 metres	25	1:1
3.	Builders line		5	1:5
4.	Mason square		5	1:5
5.	Timber	Assorted	Sufficient	
6.	Nails	Assorted	Sufficient	
7.	Claw bar		5	1:5
8.	Cement/ lime		25 bags	1:1
9.	spade		5	1:5
10.	Trowel		25	1:1
11.	Spirit level		5	1:5
12.	Plumb bob		25	1:1
13.	Wood float		25	1:1
14.	Bar bender		5	1:5
15.	BRC Mesh/ Reinforcement bars		5	1:5
16.	Dpc		1 roll	1:5
17.	Ballast		10Ton	1:5
18.	Sand		10Ton	1:5
19.	Concrete mixer		1	1:25
20.	Tamping rod/ poker		1	1:25

	vibrator			
21.	Prefabricated building casements		sufficient	
22.	Batching boxes		5	1:5
23.	Hoop Iron		1 roll	1:5
24.	Binding wire		1 roll	1:5
B	Learning Facilities & Infrastructure			
25.	LectureRoom	(9* 8 sq. metres)	1	1:25
26.	Workshop/ construction site		1	1:25
27.	Internet Connection	System	1	1:25
28.	Whiteboard	Glass, melamine, porcelain	1	1:25

BUILDING WORKS FINISHES II

UNIT CODE: 0732 453 17A

TVET CDACC UNIT CODE: CON/CU/BLD/CR/05/5/MA

UNIT DURATION: 80 Hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: Perform Building Finishes works II

Unit Description

This unit specifies the competencies required to perform building finishes. It involves laying tiles, laying terrazzo and applying Paint. These are tasks and activities carried out at almost the tail end of construction.

Summary of Learning Outcomes

S/NO	LEARNING OUTCOME	DURATION(HOURS)
1	Lay tiles on surfaces	26
2	Lay terrazzo on surfaces	36
3	Apply Paint	18
	TOTAL	80

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Lay tiles on surfaces	1.1 PPEs 1.2 <i>Tiling tools and equipment</i> 1.3 Tiling Surface screeding 1.4 <i>Tiling materials</i> 1.5 Tile setting	• Practical • Projects • POE evaluation • Third party reports • Written tests

	1.6 Tile cutting. 1.7 Tiles fixing 1.8 Tile joint 1.9 Tiled surface cleaning	
2. Lay terrazzo on surfaces	2.1 PPEs 2.2 <i>Terrazzo tools and equipment</i> 5.1 Terrazzo Surface screeding 2.3 <i>Terrazzo materials</i> 2.4 Terrazzo dividing strips 2.5 Mixing Terrazzo chips. 2.6 Green terrazzo mixture 2.7 Green terrazzo mixture compacting 2.8 curing 2.9 Terrazzo surface grounding 2.10 Polishing Terrazzo surface	• Practical • Projects • POE evaluation • Third party reports • Written tests
3. Apply Paint	3.1 PPEs 3.1.1 Types 3.1.2 uses 3.2 Cordoning off methods 3.2.1 Physical barrier 3.2.2 Temporary barriers 3.2.3 Safety signage 3.2.4 Masking and taping 3.2.5 Work site zoning 3.3 Working platform methods. 3.3.1 scaffolding 3.4 Surface preparation 3.4.1 Filler 3.4.2 Sanding 3.4.3 Scrapping existing paint 3.5 Setting out of painting area.	• Written Assessments • Reports • Written Tests • Practical Assessments • Observation

	3.6 Paint mixing 3.6.1 Manual mixing 3.6.2 Mechanical mixing 3.6.3 Computerised mixing 3.7 Application paint 3.7.1 Brush painting 3.7.2 Roller painting 3.7.3 Spray painting 3.8 Painting equipment maintenance	
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Suggested Methods of Instruction

- Practical
- Projects
- Demonstrations
- Group Discussion
- Direct Instruction

Recommended Resources for 25 Trainees

S/No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Trainee)
A	Learning Materials			
1.	Desktop computers/laptops	Any model	5	1:5
2.	Tape measures	5 metres	25	1:1
3.	Builders line		5	1:5
4.	Mason square		5	1:5
5.	Cement/ lime		25 bags	1:1
6.	spade		5	1:5
7.	Trowel		25	1:1
8.	Spirit level		5	1:5
9.	Plumb bob		25	1:1
10.	Wood float		25	1:1
11.	Steel float		25	1:1
12.	Concrete chisel		25	1:1

13.	Straight edge		5	1:5
14.	Sand		10Ton	1:5
15.	Tiles		10 packets	1:5
16.	Tile cutter		5	1:5
17.	Tile spacer		5 packets	1:5
18.	Concrete mixer		1	1:25
19.	Tamping rod/ poker vibrator		1	1:25
20.	Terrazzo chippings		10 packets	1:5
21.	Surface grinder		1	1:25
22.	Wire brush		5	1:5
23.	Keying rod		5	1:5
24.	Batching boxes		5	1:5
B	Learning Facilities & Infrastructure			
25.	LectureRoom	(9* 8 sq. metres)	1	1:25
26.	Workshop/ construction site		1	1:25
27.	Internet Connection	System	1	1:25
28.	Whiteboard	Glass, melamine, porcelain	1	1:25